**BATCH NO:16**

# SMS BASED VACCINATION REMINDER

*Community Service Project report submitted*
*in partial fulfillment of the requirement for award of the degree of*

**Bachelor of Technology**
**in**
**Artificial Intelligence and Machine Learning**

**By**


**B AKSHAYA** (24UEAM0010) **(VTU28537)**
**P VENKAT** (24UEAM0070) **(VTU29111)**

*Under the guidance of*
*Dr.M.Misba,Associate Professor,*
*B.E.Phd*

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING**

**SCHOOL OF COMPUTING**

**VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF SCIENCE & TECHNOLOGY**

**(Deemed to be University Estd u/s 3 of UGC Act, 1956)**
**Accredited by NAAC with A++ Grade**
**CHENNAI 600 062, TAMILNADU, INDIA**


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**April, 2026**

# CERTIFICATE

It is certified that the work contained in the project report titled ”SMS BASED VACCINATION REMINDER” by ”B AKSHAYA (24UEAM0010), P VENKAT (24UEAM0070)” has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

**Signature of Supervisor**
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**April, 2026**

**Signature of Head of the Department**
**Dr. S. Alex David**
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**Institute of Science & Technology**
**April , 2026**

**Signature of the Dean**
**Dr. S P. Chokkalingam**
**Professor & Dean**
**School of Computing**
**Vel Tech Rangarajan Dr. Sagunthala R&D**
**Institute of Science & Technology**
**April, 2026**

# DECLARATION

We hereby declare that we have completed the mandatory community service project in the stipulated time period in (Kaluvaya) under the guidance of our Project supervisor

(Signature)

(B AKSHAYA)

Date: / /

(Signature)

(P VENKAT)

Date: / /

# APPROVAL SHEET

This project report entitled (SMS BASED VACCINATION REMINDER) by (B AKSHAYA (24UEAM0010), (P VENKAT (24UEAM0070) is approved for the degree of B.Tech in Artificial Intelligence and Machine Learning.

**Examiners** **Supervisor**

Dr.M.Misba,Associate Professor,B.E.Phd,.

**Date: / /**

**Place:**

# ACKNOWLEDGEMENT

We express our deepest gratitude to our **Honorable Founder Chancellor and President Col. Prof. Dr. R. RANGARAJAN B.E. (Electrical), B.E. (Mechanical), M.S (Automobile), D.Sc., and Foundress President Dr. R. SAGUNTHALA RANGARAJAN M.B.B.S. Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology,** for their blessings.

We express our sincere thanks to our respected Chairperson and Managing Trustee **Dr. RANGARAJAN MAHALAKSHMI KISHORE, B.E., M.B.A. Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology,** for her blessings.

We are very much grateful to our beloved **Vice Chancellor Prof. Dr.RAJAT GUPTA , M.Tech., Ph.D.,** for providing us with an environment to complete our project successfully.

We record indebtedness to our **Professor & Dean, School of Computing, Dr.SP.CHOKKALINGAM, M.Tech., Ph.D., Professor & Associate Dean, Dr.V.DHILIP KUMAR, M.E., Ph.D., Professor & Assistant Dean, Dr. R. PARTHASARATHY, M.E., Ph.D.,** for their immense care and encouragement towards us throughout the course of this project.

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A special thanks to our **Project Coordinators Dr B. PRABHU SHANKAR Ph.D.,** for his valuable guidance and support throughout the course of the project.

We thank our department faculty, supporting staff and friends for their support and guidance to complete this project.

**B AKSHAYA (24UEAM0010)**
**P VENKAT (24UEAM0070)**

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# LIST OF ACRONYMS AND ABBREVIATIONS

**SMS** Short Message Service

**WHO** World Health Organization

**UNICEF** United Nations Children's Fund

**CDC** Centers for Disease Control and Prevention

**UIP** Universal Immunization Programme

**NHM** National Health Mission

**ICT** Information and Communication Technology

**AI** Artificial Intelligence

**ML** Machine Learning

**GPS** Global Positioning System

**PHC** Primary Health Centre

**ASHA** Accredited Social Health Activist

**OPD** Out Patient Department

**API** Application Programming Interface

**DBMS** Database Management System

**OTP** One Time Password

**UI** User Interface

**UX** User Experience

**URL** Uniform Resource Locator

**IoT** Internet of Things

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# Chapter 1

# EXECUTIVE SUMMARY

SMS-based vaccination reminder systems are designed to improve immunization coverage by sending timely alerts to individuals about upcoming or missed vaccine doses. These systems typically use mobile phone networks to deliver simple text messages that notify parents, patients, or caregivers about vaccination schedules, dates, and nearby healthcare facilities. In regions where smartphone or internet access is limited, SMS serves as an effective and low-cost communication tool, ensuring that even rural populations receive critical health information. By automating reminders, healthcare providers can reduce the chances of missed vaccinations and ensure better adherence to immunization programs.

Such systems also help bridge the communication gap between healthcare institutions and the public. They can be integrated with health databases to track patient records and generate personalized reminders based on age, vaccine type, and schedule. In addition to reminders, SMS systems can also provide educational messages about the importance of vaccines, possible side effects, and follow-up care. Overall, SMS-based vaccination reminders contribute to improved public health outcomes by increasing awareness, reducing forgetfulness, and supporting timely vaccination practices.

The SMS-Based Reminder System for Community Vaccination and Health Check-up is designed to improve public health services by ensuring timely communication between healthcare providers and community members. In many communities, especially in rural and semi-urban areas, people often miss vaccination schedules and regular health check-ups due to lack of awareness, forgetfulness, or poor communication systems.

### Goals and Objectives

This project proposes an automated SMS reminder system that sends notifications to individuals regarding upcoming vaccination dates and health check-up appointments. The system collects and stores patient details, schedules, and contact information in a centralized database. Using a scheduling mechanism, reminders are generated and sent through an SMS gateway to ensure that users receive timely alerts.

### Intended Impact

The implementation of SMS-based vaccination reminders has a significant positive impact on public health.The proposed SMS-based vaccination reminder system is intended to significantly improve timely immunization coverage in rural areas like Kaluvaya village. By sending automated reminders to parents and guardians, the system helps reduce missed or delayed vaccinations caused by forgetfulness or lack of awareness. This ensures that children receive vaccines on time, thereby lowering the risk of preventable diseases and improving overall community health.

In addition, the system aims to enhance communication between healthcare providers and the com-

munity. It reduces the burden on healthcare workers by minimizing manual follow-ups and improving record management. The use of simple SMS technology makes the solution cost-effective, scalable, and accessible, even in areas with limited internet connectivity. Overall, the system contributes to better healthcare delivery, increased awareness, and stronger public health outcomes.

### Beneficiaries

The primary beneficiaries are infants and children, along with parents and healthcare providers.The proposed SMS-based vaccination reminder system benefits several groups within the community. Children benefit by receiving vaccinations on time, ensuring better protection against diseases. Parents and guardians gain timely reminders and clear information, reducing confusion and missed appointments. Healthcare workers benefit from reduced workload and improved efficiency in managing vaccination schedules. Overall, the community experiences improved health outcomes, and the healthcare system benefits from increased immunization coverage and better service delivery.

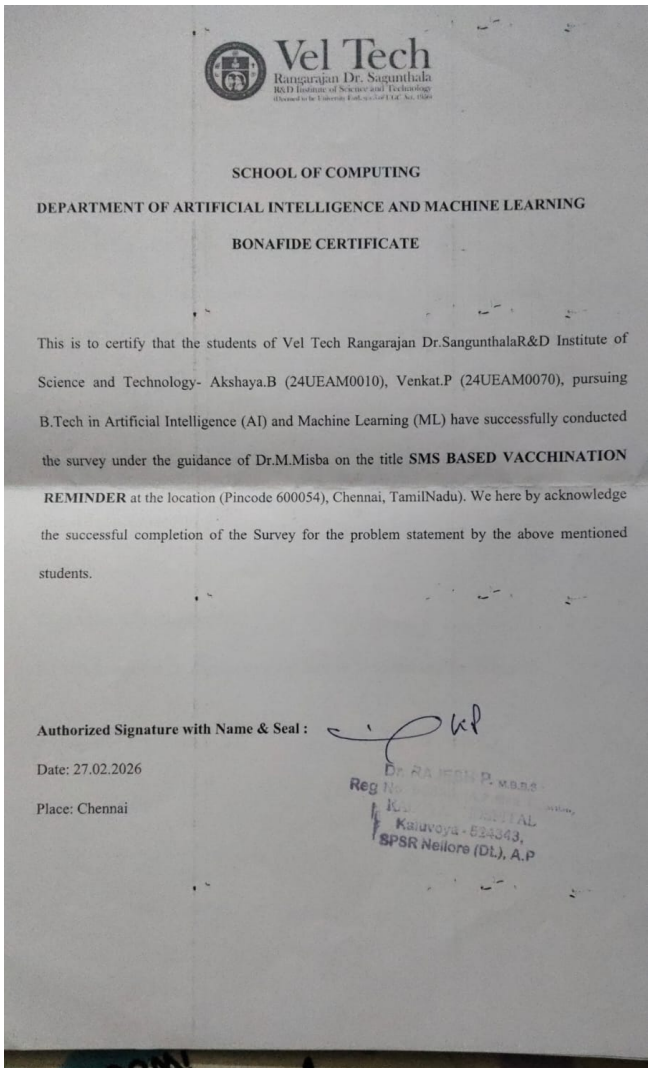
### Social Issue

A major social issue is the lack of awareness about vaccination schedules and their importance.The primary social issue identified in this project is the low and inconsistent vaccination coverage in rural areas such as Kaluvaya village. Many children miss or delay essential vaccinations due to lack of awareness about schedules, poor communication between healthcare providers and the community, and simple forgetfulness. These challenges increase the risk of preventable diseases and negatively impact overall public health. Additionally, healthcare workers face difficulties in manually tracking and reminding all beneficiaries, which leads to gaps in timely immunization.

# Chapter 2

# OVERVIEW OF THE COMMUNITY

## 2.1 Certificate from the Office of Community



Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology

SCHOOL OF COMPUTING

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

BONAFIDE CERTIFICATE

This is to certify that the students of Vel Tech Rangarajan Dr.SangunthalaR&D Institute of Science and Technology- Akshaya.B (24UEAM0010), Venkat.P (24UEAM0070), pursuing B.Tech in Artificial Intelligence (AI) and Machine Learning (ML) have successfully conducted the survey under the guidance of Dr.M.Misba on the title **SMS BASED VACCHINATION REMINDER** at the location (Pincode 600054), Chennai, TamilNadu). We here by acknowledge the successful completion of the Survey for the problem statement by the above mentioned students.

**Authorized Signature with Name & Seal :**

Date: 27.02.2026

Place: Chennai

Kaluvoya - 524343,
SPSR Nellore (Dt.), A.P

Figure 2.1: Certificate from the Office of Community

# Chapter 3

# ACTIVITY LOG OUTCOME

## 3.1 Project Activity Log

As part of the project, a field survey was conducted in Kaluvaya to understand the current vaccination practices and challenges faced by the local population. The team interacted with parents, caregivers, and community members to gather information about their awareness of vaccination schedules, accessibility to healthcare services, and common reasons for missed immunizations. The survey also focused on identifying whether people had access to mobile phones and their willingness to receive SMS-based reminders for vaccinations.

The collected data revealed key issues such as lack of awareness, forgetfulness, and limited communication from healthcare providers. Many respondents expressed interest in receiving timely SMS alerts to help them keep track of vaccination dates. This activity helped the team gain practical insights into real-world problems and supported the need for implementing an SMS-based vaccination reminder system to improve immunization coverage in the area.

| Date Range | Time | Location | Activity Description |
|---|---|---|---|
| Day 1 | 2-4hrs | Home | Discussed and finalized the project topic “SMS-Based Vaccination Reminder System.” |
| Day 2 | 2-3hrs | Home | Collected basic information about vaccination systems and their importance. |
| Day 3 | 3hrs | Lab | Studied problems like missed vaccinations, lack of awareness, and communication gaps. |
| Day 4 | 1hrs | Lab | Defined project objectives and scope based on identified issues. |
| Day 5 | 2hrs | Home | Prepared general survey questions related to vaccination awareness and reminders. |
| Day 6 | 2.5hrs | Lab | Reviewed and refined the questionnaire to make it simple and clear. |
| Day 7 | 1hr | Kaluvaya | Conducted field survey in Kaluvaya village. |
| Day 8 | 1.5hrs | Kaluvaya | Interacted with parents and caregivers to collect responses. |
| Day 9 | 3hrs | Home | Recorded and organized survey data systematically. |
| Day 10 | 1hr | Home | Analyzed responses to identify key findings and patterns. |
| Day 11 | 3hrs | Home | Summarized survey results and identified major issues. |
| Day 12 | 2hrs | Home | Drafted content for problem statement, goals, and social issues. |
| Day 13 | 3hrs | Home | Started preparing the project report in Overleaf. |
| Day 14 | 2-3hrs | Home | Added tables, activity logs, and formatted the document. |
| Day 15 | 3hrs | Home | Reviewed content and corrected formatting/alignment issues. |
| Day 16 | 4hrs | Home | Final editing, proofreading, and prepared for submission. |

Table 3.1: Detailed Project Activity Log with Date,Time,Location

## 3.2 Project Outcome

The project resulted in a comprehensive understanding of the challenges associated with vaccination awareness and adherence to immunization schedules. Through the field survey conducted in Kaluvaya, primary data was collected from parents and caregivers, which helped in identifying key issues such as lack of awareness about vaccination schedules, forgetfulness, and insufficient communication from healthcare providers. The analysis of this data provided meaningful insights into the real-world gaps in the existing system. Although a working prototype was not developed, the project successfully proposed an SMS-based vaccination reminder system as a practical and cost-effective solution. The final outcome includes a well-organized report that clearly presents the problem, survey findings, and the proposed approach to improve vaccination coverage and public health outcomes.

The progress of the project was monitored on a day-to-day basis by dividing tasks into smaller activities such as topic selection, survey preparation, data collection, analysis, and documentation. Each stage was completed systematically within the planned timeline. Regular reviews were conducted to ensure that the work was aligned with project objectives, and necessary corrections were made during documentation and formatting stages.

**Tracking Progress**

The project progress was systematically tracked over a period of 16 days by dividing the work into structured phases. Initially, time was allocated for topic selection, background study, and defining objectives. This was followed by survey design, field data collection, and interaction with respondents. The collected data was then organized and analyzed to extract key findings. Regular monitoring of daily activities ensured that each phase was completed within the planned schedule. Any delays or difficulties encountered during survey or documentation were addressed promptly. Progress tracking also involved reviewing the quality of collected data and making improvements in the questionnaire and documentation. This structured approach helped in maintaining consistency and achieving the project goals efficiently.

**Communication**

Communication played a crucial role in the successful completion of the project. During the survey phase, effective communication with the residents of Kaluvaya enabled the collection of accurate and meaningful data. Questions were explained clearly to ensure proper understanding among respondents, which improved the reliability of responses. Within the project work, continuous communication and discussion helped in refining ideas, identifying key issues, and making decisions regarding the structure of the report. Feedback obtained during interactions was carefully considered and incorporated into the analysis. Overall, clear and effective communication ensured smooth coordination and contributed significantly to the quality of the project.

**Documentation**

Documentation was an essential part of the project and was carried out using LaTeX in Overleaf to ensure a professional and structured presentation. All stages of the project, including introduction, problem statement, objectives, survey methodology, data analysis, activity log, and conclusions, were systematically recorded. Proper formatting techniques such as tables, headings, and alignment were

used to improve readability and organization. Special attention was given to maintaining consistency in formatting and clarity in content. Errors identified during review were corrected, and the document was refined to meet academic standards. The final documentation effectively represents the entire project work in a clear, concise, and well-structured manner, making it suitable for evaluation and submission.

The project on "SMS-Based Vaccination Reminder System" resulted in a comprehensive analysis of the current challenges associated with vaccination awareness and timely immunization. Through systematic research and a field survey conducted in Kaluvaya, the project successfully gathered primary data that reflects real-world conditions. The responses collected from parents and caregivers highlighted several critical issues, including lack of awareness about vaccination schedules, difficulty in remembering important dates, and limited communication between healthcare providers and the public. These findings clearly demonstrate the need for an effective and accessible reminder system.

One of the key outcomes of the project is the identification of SMS technology as a simple, cost-effective, and widely accessible solution to address these challenges. Since mobile phones are commonly used even in rural areas, SMS-based reminders can serve as a reliable medium to deliver timely notifications about upcoming or missed vaccinations. The project emphasizes that such a system can significantly improve adherence to vaccination schedules, reduce the chances of incomplete immunization, and ultimately contribute to better public health outcomes.

In addition to identifying the problem and proposing a solution, the project also enhanced the understanding of survey-based research methodologies. The process of designing questionnaires, conducting field surveys, and analyzing responses provided valuable practical experience. It helped in developing skills related to data collection, organization, and interpretation. The structured analysis of survey data enabled the identification of patterns and trends, which were crucial in drawing meaningful conclusions.

Another important outcome is the development of a well-organized and professionally formatted project report using LaTeX in Overleaf. The documentation includes all essential components such as introduction, problem statement, objectives, survey analysis, activity logs, and conclusions. The use of proper formatting techniques, tables, and structured sections improved the clarity and presentation of the report. This not only reflects the work carried out during the project but also demonstrates the ability to present technical information effectively.

Overall, the project successfully achieved its objectives by identifying real-world problems, analyzing them through primary data collection, and proposing a feasible solution. Even without the development of a working prototype, the project provides a strong conceptual foundation for future implementation. It highlights the potential impact of SMS-based reminder systems in improving vaccination coverage, increasing awareness, and supporting preventive healthcare practices. The outcomes of this project can be further extended and implemented in real-life scenarios to benefit communities and enhance public health systems.

# Chapter 4

# BACKGROUND SURVEY ANALYSIS FOR PROBLEM STATEMENT

To clearly understand the challenges associated with vaccination awareness and adherence, a background survey was conducted in Kaluvaya village. The primary aim of this survey was to collect real-time data from parents, caregivers, and community members regarding their knowledge of vaccination schedules, accessibility to healthcare services, and the difficulties they face in ensuring timely immunization. The survey included general questions related to awareness levels, frequency of missed vaccinations, availability of reminders, and willingness to adopt an SMS-based notification system.

The analysis of the collected data revealed several important insights. A significant number of respondents were not fully aware of the complete vaccination schedule, especially beyond the initial stages of immunization. Many parents relied on memory or informal sources for tracking vaccination dates, which often resulted in missed or delayed doses. Forgetfulness was identified as one of the major reasons for incomplete immunization. Additionally, there was a noticeable communication gap between healthcare providers and the community, as most respondents reported not receiving any prior reminders or notifications about upcoming vaccinations.

## 4.1 Background Study

Vaccination is one of the most effective public health interventions for preventing infectious diseases and reducing child mortality. Programs such as India's Universal Immunization Programme (UIP) aim to provide free vaccines to children and pregnant women. Despite these efforts, many rural areas still experience incomplete or delayed immunization due to various challenges.

One of the major issues identified in rural communities is the lack of awareness about vaccination schedules. Many parents are unaware of exact dates or intervals between doses, which leads to missed or delayed vaccinations. In addition, factors such as forgetfulness, limited access to healthcare information, and communication gaps between healthcare providers and residents further contribute to low immunization coverage In the context of vaccination, SMS reminders can play a crucial role by:

- Informing parents about upcoming vaccination dates
- Reducing missed immunization cases
- Improving communication between healthcare centers and the community

The present study focuses on understanding the feasibility of such a system through a field survey conducted in Kaluvaya village.The survey examines community awareness, mobile accessibility.

## 4.2 Background Study

Immunization plays a crucial role in preventing life-threatening diseases, especially among infants and young children. Despite various government initiatives and vaccination programs, missed or delayed vaccinations remain a common issue in rural and semi-urban areas. One of the major reasons for this problem is lack of awareness, poor tracking of vaccination schedules, and limited access to timely healthcare reminders. As a result, there is a strong need for an efficient and low-cost communication system that can help improve vaccination compliance.

With the rapid growth of mobile phone usage, Short Message Service (SMS) has emerged as one of the most effective communication tools, especially in rural regions where smartphone and internet penetration may be limited. SMS does not require internet connectivity and can reach users instantly, making it a reliable medium for healthcare communication. This has led to the adoption of SMS-based systems in various healthcare applications such as appointment reminders, medication alerts, and public health awareness campaigns.

The concept of Mobile Health (mHealth) focuses on the use of mobile technologies to support healthcare services and information delivery. Many studies have shown that SMS reminders significantly improve patient attendance and adherence to medical schedules. In the context of vaccination programs, SMS-based reminder systems help ensure that parents or guardians are informed about upcoming vaccination dates, types of vaccines, and healthcare center details. This reduces the chances of missed appointments and improves overall immunization coverage.

Several healthcare organizations, including the World Health Organization (WHO) and UNICEF, have highlighted the importance of digital communication tools in strengthening immunization programs. These systems are particularly useful in developing countries where manual tracking is inefficient and healthcare resources are limited. By automating reminders, healthcare workers can reduce their workload while ensuring better patient engagement.

Therefore, the development of an SMS-based vaccination reminder system provides a simple, cost-effective, and scalable solution to improve public health outcomes. It bridges the communication gap between healthcare providers and beneficiaries, ensuring timely vaccination and regular health checkups. This project aims to contribute to the improvement of healthcare delivery systems by leveraging mobile technology for better awareness and compliance.

## 4.3 Literature Review

| S.No | Author / Title | Year | Findings |
|---|---|---|---|
| 1 | World Health Organization (WHO), Immunization Agenda 2030 | 2020 | Emphasized the importance of timely vaccination and the use of digital communication tools to improve global immunization coverage. |
| 2 | UNICEF, Mobile Communication for Child Health Programs | 2021 | Reported that SMS reminders help parents remember child vaccination schedules and improve attendance at health centers. |
| 3 | K. Patel and R. Sharma, Role of Mobile Technology in Healthcare Reminder Systems | 2022 | Concluded that SMS reminder systems are cost-effective and highly useful in rural healthcare communication. |
| 4 | S. Reddy and P. Kumar, SMS-Based Reminder Systems for Improving Immunization Coverage in Rural Areas | 2021 | Found that automated reminders significantly reduced missed vaccination appointments in rural communities. |
| 5 | A. Singh et al., Impact of Mobile Health Interventions on Vaccination Compliance | 2023 | Showed that mobile-based reminders improved vaccination compliance and increased public health awareness. |
| 6 | World Bank, Digital Health and Rural Healthcare Access Report | 2022 | Highlighted that digital health systems can bridge communication gaps in underserved villages. |
| 7 | National Health Mission (NHM), Maternal and Child Health Services Report | 2021 | Recommended using mobile technology to improve maternal and child vaccination follow-up services. |
| 8 | M. Gupta and T. Joseph, Role of ICT in Rural Health Monitoring Systems | 2021 | Explained that ICT tools such as SMS alerts improve monitoring, reporting, and healthcare service delivery. |

| S.No | Author / Title | Year | Findings |
|---|---|---|---|
| 9 | I. Istepanian et al., m-Health: Emerging Mobile Health Systems | 2006 | Discussed the development of mobile health systems and their role in remote healthcare monitoring and communication. |
| 10 | HealthIT.gov, Electronic Health Records and Patient Reminder Systems | 2022 | Highlighted that integrating reminder systems with electronic health records improves patient tracking and healthcare efficiency. |
| 11 | Twilio Inc., SMS API for Healthcare Communication Systems | 2023 | Explained how SMS gateway APIs enable scalable and reliable automated healthcare notification systems. |
| 12 | Eric Topol, Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again | 2019 | Emphasized the role of AI in improving healthcare systems, predictive analysis, and patient communication. |
| 13 | 3GPP, Short Message Service (SMS) Technical Specification Standards | 2020 | Defined global technical standards for SMS communication used in telecom-based alert systems. |
| 14 | L. Free et al., Effectiveness of Mobile Health Technologies to Improve Healthcare Service Delivery | 2013 | Demonstrated that mobile health interventions significantly improve patient follow-up, treatment adherence, and healthcare service delivery using mobile communication tools. |
| 15 | K. Stockwell et al., Text Message Reminders for Vaccination Coverage Improvement | 2012 | Found that SMS-based reminder systems significantly increase vaccination rates and reduce missed immunization appointments, especially in rural and underserved populations. |

Table 4.1: Literature Review on SMS-Based Vaccination Reminder System

## 4.4 Survey Analysis with report

A survey was conducted in Kaluvaya village to assess awareness and challenges related to vaccination and to evaluate the feasibility of an SMS-based reminder system. The results showed that a large number of respondents have limited knowledge of vaccination schedules and often depend on healthcare workers for information. Several participants reported missing vaccination dates, mainly due to forgetfulness, lack of clear communication, and uncertainty about timings. This indicates that delays in vaccination are generally unintentional and caused by gaps in awareness rather than resistance.

The survey also revealed that most households have access to mobile phones, making SMS a practical and accessible communication method. A majority of respondents expressed willingness to receive SMS reminders, preferably in simple and local language. Healthcare staff also supported the idea, stating that it could reduce their workload and improve tracking of patients. Overall, the analysis suggests that an SMS-based vaccination reminder system is a feasible and effective solution to improve timely immunization in rural areas.

A survey was conducted in rural areas to understand the awareness, acceptance, and effectiveness of vaccination reminder systems among parents and guardians. The primary objective of the survey was to analyze the current challenges faced in maintaining vaccination schedules and to evaluate the need for an SMS-based reminder system.

The survey included respondents from different age groups, educational backgrounds, and socio-economic conditions. Most participants reported that they often forget vaccination dates due to busy schedules, lack of reminders, or insufficient communication from healthcare centers. It was observed that manual tracking methods are not reliable, especially in rural regions where access to healthcare information is limited.

A significant portion of the respondents expressed that they rely mainly on healthcare workers or local announcements for vaccination updates. However, delays and miscommunication were commonly reported issues. Many participants also mentioned that they would prefer receiving automated SMS reminders as they are simple, timely, and do not require internet access or smartphones.

The survey analysis further revealed that awareness about digital health solutions is gradually increasing, but adoption is still limited due to lack of infrastructure and training. Despite this, there was a strong positive response toward the implementation of mobile-based reminder systems. Parents and guardians believed that such systems would help reduce missed vaccinations and improve child healthcare management.

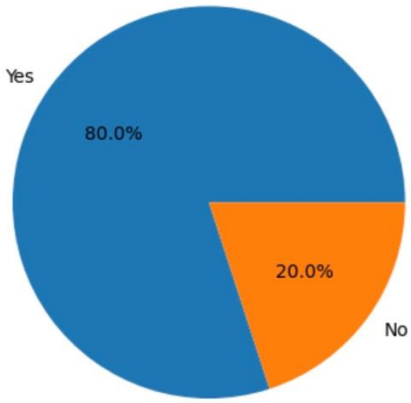



Figure 4.1: figure 4.1

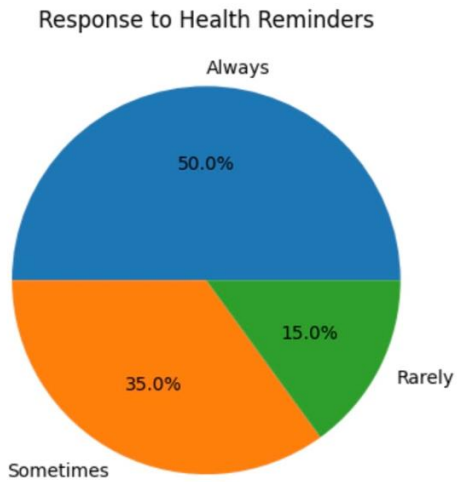



Figure 4.1: Survey Analysis Result

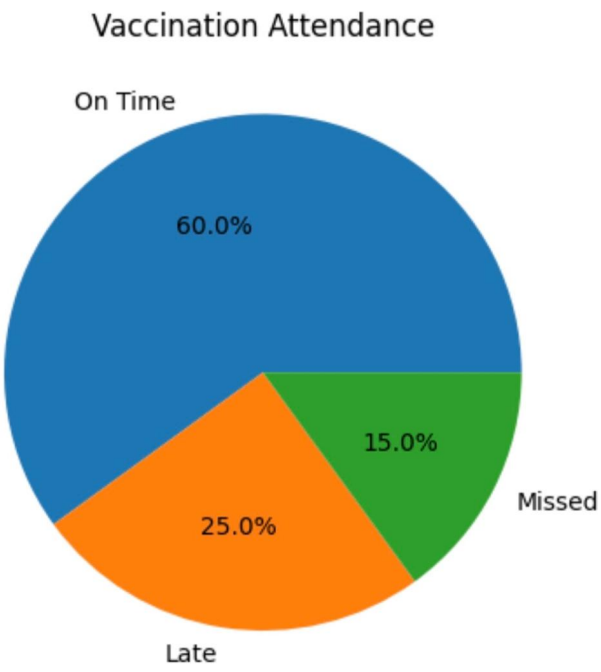



Figure 4.2: Survey Analysis Result

Overall, the survey findings clearly indicate the necessity of an automated SMS-based vaccination reminder system. It highlights the gap in the current healthcare communication process and supports the need for a simple, cost-effective, and scalable solution to improve immunization coverage and ensure better public health outcomes.

## 4.5 Geotagged Photos and Details

Geotagged photos were collected during the field survey in Kaluvaya village to provide visual evidence of the study and to record the exact location of the survey activities. These photos include embedded location data such as latitude and longitude, which helps in verifying the authenticity and accuracy of the fieldwork conducted.

Figure 4.3: Conducting Survey

Figure 4.4: Asking Generic Questions

Figure 4.5: Survey Conducted Hospital

Figure 4.6: Asking Generic Questio

Figure 4.7: Conducting Survey

## 4.6 Society Relevant Problem Identification

During the field survey conducted in Kaluvaya village, several issues related to vaccination awareness and accessibility were identified. A major problem observed was the lack of awareness among parents and guardians regarding vaccination schedules, which often leads to missed or delayed immunization. Many respondents were not aware of exact dates or follow-up doses and depended entirely on healthcare workers for updates.

Another significant issue is poor communication between healthcare providers and the community. Due to limited manpower and manual tracking methods, healthcare staff find it difficult to remind every individual about upcoming vaccinations. Additionally, forgetfulness and lack of timely information were common reasons reported for missed vaccinations. Despite the widespread availability of mobile phones in the village, they are not being effectively used for healthcare communication.

These identified problems highlight a clear gap in the existing system, where basic awareness and communication mechanisms are insufficient. Addressing these issues is essential to improve vaccination coverage and overall public health in rural areas.

## 4.7 Development of Problem solution

Based on the issues identified during the survey in Kaluvaya village, a conceptual solution is proposed in the form of an SMS-based vaccination reminder system. The solution focuses on addressing key problems such as lack of awareness, forgetfulness, and communication gaps between healthcare providers and the community.

The proposed system would work by collecting basic details of beneficiaries, including name, mobile number, and vaccination schedule, at the local healthcare center. Using this information, automated SMS reminders can be sent to parents or guardians before the due date of each vaccination. The messages would be designed in a simple and local language to ensure easy understanding. This approach utilizes existing mobile phone accessibility in rural areas, making it a cost-effective and practical solution.

Additionally, the system can assist healthcare workers by reducing the burden of manual tracking and follow-ups. It ensures timely communication, improves record management, and helps in increasing vaccination adherence. Although this project does not include a working prototype, the proposed solution demonstrates how a simple technological intervention can effectively address a real-world societal problem and enhance public health outcomes. Based on the issues identified during the survey in Kaluvaya village and Morai, a conceptual solution is proposed in the form of an SMS-based vaccination reminder system. The survey findings highlighted key challenges such as lack of awareness about vaccination schedules, frequent forgetfulness among parents or guardians, and communication gaps between healthcare providers and the community. These issues collectively contribute to missed or delayed vaccinations, which can negatively impact child health and overall community well-being.

The proposed system aims to address these challenges by leveraging the widespread availability of mobile phones in rural areas. It would begin with the collection of basic beneficiary details, including name, child information, mobile number, and vaccination schedule, at the local healthcare center. This data would be stored in a structured format and used to track upcoming vaccination dates. Based on this information, automated SMS reminders would be generated and sent to parents or guardians prior to each scheduled vaccination.

The reminder messages would be designed in simple and local language to ensure clarity and easy understanding for all users, regardless of their literacy level. These messages may include details such as the date of vaccination, type of vaccine, and the location of the healthcare facility. By providing timely and clear notifications, the system helps reduce dependency on memory and ensures that beneficiaries are well-informed in advance.

1. Identified the problem of missed vaccination schedules due to lack of awareness and forgetfulness among villagers.
2. Conducted a survey in Kaluvaya village to understand vaccination challenges and mobile phone usage.
3. Collected data showing that most people own mobile phones and are willing to receive SMS reminders.
4. Designed an SMS-based vaccination reminder system to send alerts before scheduled vaccination dates.
5. Developed a database to store beneficiary details such as name, mobile number, age, and vaccination schedule.
6. Created an automated messaging module to send reminder messages at regular intervals.
7. Included local language support such as Telugu for better understanding by villagers.
8. Integrated health worker support for updating records and monitoring vaccination status.
9. Tested the system for message delivery, accuracy, and user response.
10. Proposed implementation in Kaluvaya village to improve vaccination coverage and public health awareness.

## 4.8 Block Diagram

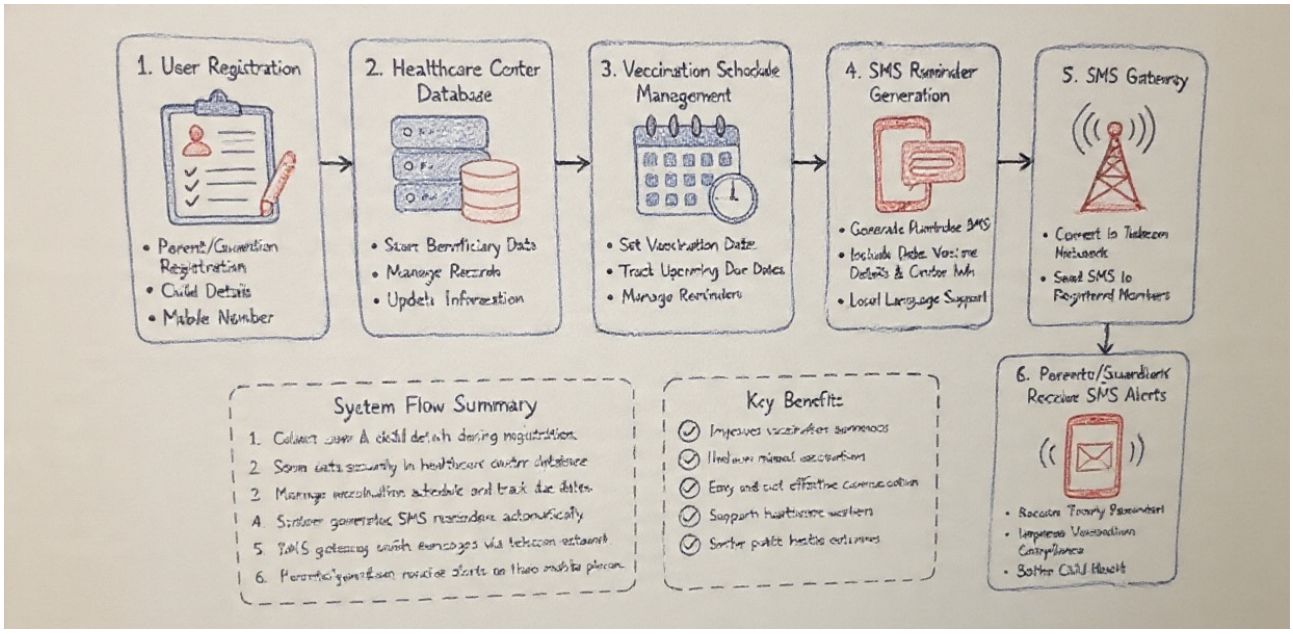



Figure 4.8: SMS Based Vaccination Reminder

The above diagram represents the working of an SMS-based vaccination reminder system designed to improve immunization coverage in rural areas. The process begins with user registration, where details such as parent/guardian name, child information, and mobile number are collected at the healthcare center. This data is then stored in a centralized database, which maintains records of beneficiaries and their vaccination history.

Next, the system manages the vaccination schedule by tracking due dates and upcoming immunizations. Based on this schedule, the system generates automated SMS reminders containing important details such as vaccination date, type of vaccine, and healthcare center information. These messages are sent through an SMS gateway, which connects to the telecom network and delivers alerts to registered mobile numbers.

Finally, parents or guardians receive timely SMS notifications, helping them remember vaccination dates and ensuring better compliance. Overall, the system provides a simple, cost-effective, and efficient solution to reduce missed vaccinations and improve public health outcomes. In addition to sending vaccination reminders, the system can also support periodic health checkup notifications for children and elderly individuals. These reminders help ensure that users do not miss important medical appointments such as routine physical examinations, growth monitoring, and preventive screenings. By maintaining continuous communication with beneficiaries, the system promotes proactive healthcare practices and reduces the risk of delayed diagnosis or treatment.

Furthermore, the system is designed to be scalable and adaptable to large populations across different geographical regions. Since it operates through SMS technology, it does not require smartphones or internet connectivity, making it highly suitable for rural and remote areas. The centralized database allows healthcare authorities to efficiently manage large datasets and monitor vaccination coverage in real time. This improves decision-making and helps government health programs identify areas with low immunization rates, enabling targeted interventions.

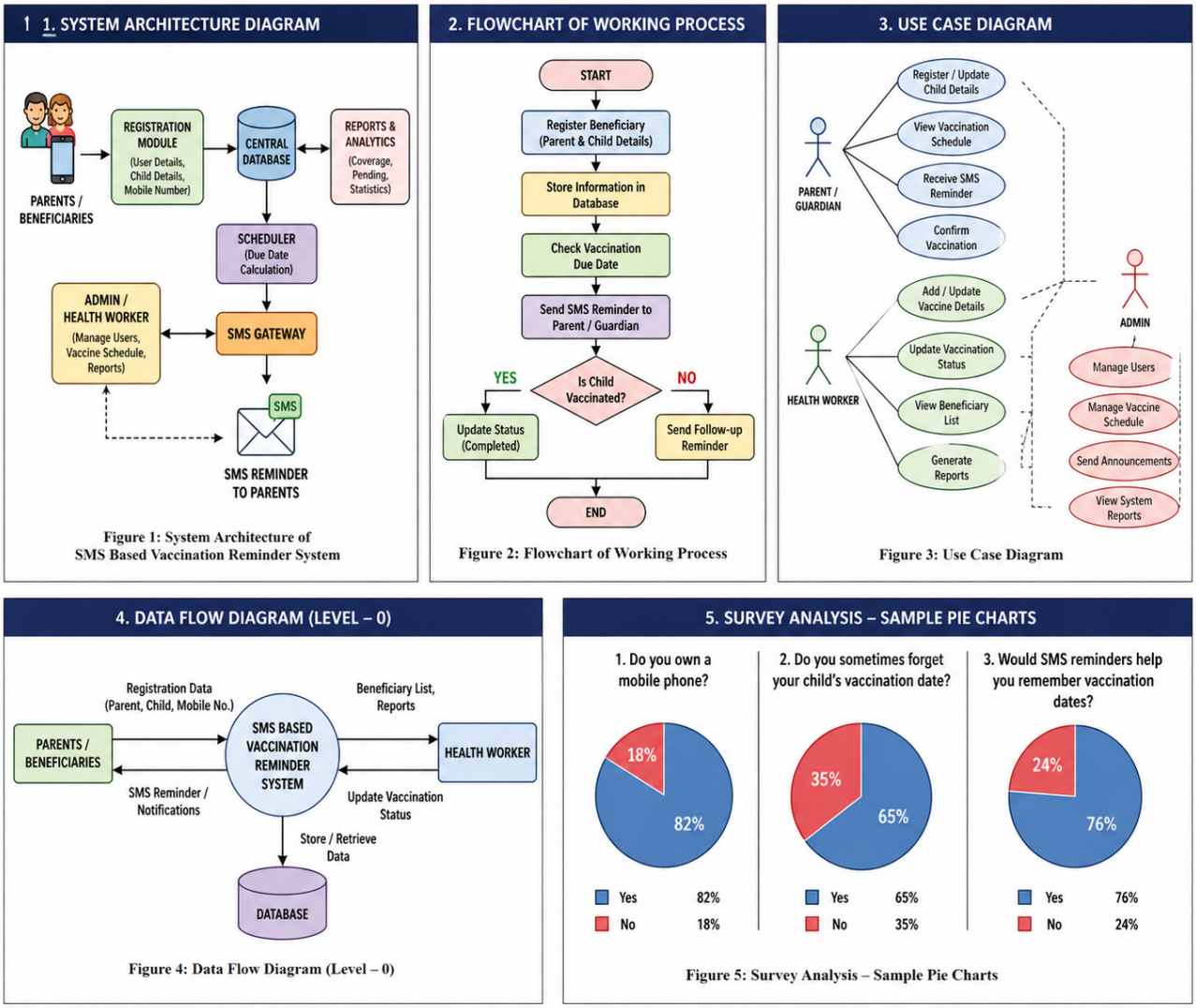



Figure 4.9: Comprehensive System Diagram for SMS-Based Vaccination Reminder System

## Section A: Demographics

1. Age Group
    - Below 18
    - 18–30
    - 31–45
    - 46–60
    - Above 60
2. Gender
    - Male
    - Female
    - Prefer not to say
3. Education Level
    - No formal education
    - School level
    - Undergraduate
    - Postgraduate
4. Occupation
    - Student
    - Employed
    - Self-employed
    - Unemployed
5. Location
    - Urban
    - Semi-urban
    - Rural

## Section B: Mobile Usage

1. Do you own a mobile phone?
    - Yes
    - No
2. Type of phone used
    - Basic phone
    - Smartphone
3. How often do you check SMS?
    - Frequently
    - Occasionally
    - Rarely
4. Can you read SMS messages?
    - Yes
    - No
5. Preferred language for SMS
    - English
    - Regional language
    - Both

## Section C: Vaccination Awareness

1. Are you aware of vaccination schedules?
    - Yes
    - No
2. Source of vaccination information
    - Hospital
    - Government campaigns
    - Media

- Family/Friends

3. Have you ever missed a vaccination?
    - Yes
    - No
4. Reason for missing vaccination
    - Forgot
    - Busy schedule
    - Lack of awareness
    - Other
5. Do you keep vaccination records?
    - Yes
    - No

## Section D: SMS Reminder System

1. Would SMS reminders be helpful?
    - Yes
    - No
2. Preferred reminder timing
    - Same day
    - 1 day before
    - 1 week before
3. Number of reminders preferred
    - One
    - Two
    - Multiple
4. Would you trust SMS reminders from healthcare providers?
    - Yes

- No

5. Do you prefer SMS over phone calls?
    - Yes
    - No

## Section E: Usability & Preferences

1. Preferred SMS content style
    - Short and simple
    - Detailed
2. Would you like vaccination location included in SMS?
    - Yes
    - No
3. Would you prefer reminders for family members too?
    - Yes
    - No
4. Should SMS include health tips?
    - Yes
    - No
5. Do you find SMS intrusive?
    - Yes
    - No

## Section F: Effectiveness & Feedback

1. Do reminders increase likelihood of vaccination?
    - Yes
    - No
2. Would you recommend this system?

- Yes
- No

3. Any privacy concerns with SMS reminders?
   - Yes
   - No
4. Would you register for this service?
   - Yes
   - No
5. Overall satisfaction with SMS reminder concept
   - Very satisfied
   - Satisfied
   - Neutral
   - Unsatisfied

## 4.9 FUTURE SCOPE

The proposed SMS-Based Vaccination Reminder System has significant potential for future enhancement and large-scale implementation. Although the current project focuses on sending reminder messages for vaccination schedules, the system can be further expanded by integrating advanced technologies and additional healthcare services. Future improvements can make the system more efficient, user-friendly, and beneficial to a larger population.

One major enhancement can be the integration of voice call reminders in addition to SMS alerts. This would be highly useful for illiterate users or elderly people who may have difficulty reading text messages. Multi-language support can also be improved by including more regional languages to ensure better communication across different communities.

The system can also be connected with mobile applications that allow users to view vaccination history, upcoming schedules, nearby healthcare centers, and digital health records. A dedicated admin dashboard for healthcare workers can be developed to monitor beneficiaries, track missed vaccinations, and generate reports for analysis.

In the future, Artificial Intelligence can be integrated to provide personalized reminders, health suggestions, and chatbot assistance for common vaccination-related queries. GPS-based services may also be added to guide users to the nearest vaccination center or hospital during emergencies.

Another important scope is the expansion of the system beyond vaccinations. It can be used for reminders related to regular health check-ups, medicine schedules, pregnancy care, blood donation camps, and public health awareness campaigns. With proper government support and technical development, the system can be implemented in villages, towns, and cities on a wider scale.

Overall, the future scope of this project is broad and promising. By combining communication technology with healthcare services, the system can play an important role in improving preventive healthcare, increasing awareness, and building healthier communities.

The proposed SMS-based vaccination and monthly health checkup reminder system can be further enhanced in several ways to improve its efficiency, scalability, and user experience.

The SMS-based vaccination and monthly health checkup reminder system has significant potential for further enhancement and expansion in the field of digital healthcare. With the increasing adoption of mobile technologies and digital health platforms, this system can evolve into a more advanced, intelligent, and integrated healthcare communication solution.

In the future, the system can be enhanced by integrating Artificial Intelligence (AI) and Machine Learning (ML) techniques to analyze patient data and predict missed vaccinations or irregular health checkups. By studying user response patterns, the system can automatically adjust reminder timing

and frequency to improve effectiveness and user engagement. This predictive capability can help healthcare providers identify high-risk individuals and take preventive actions in advance.

Another major improvement can be the integration of the system with national and state-level healthcare databases. This would allow real-time synchronization of vaccination records, patient history, and appointment schedules. Such integration will reduce manual data entry errors and ensure that users always receive accurate and up-to-date health reminders.

The system can also be extended beyond SMS by incorporating voice call reminders and interactive voice response (IVR) systems. This will be especially useful for illiterate users or those who are not comfortable with text messages. Voice-based communication in regional languages can significantly increase accessibility in rural and remote areas.

Furthermore, the development of a companion mobile application can provide additional features such as digital health records, online appointment booking, vaccination tracking dashboards, and direct communication with healthcare professionals. This hybrid approach of SMS and mobile application will ensure broader coverage across both basic phone users and smartphone users.

The future system can also leverage Internet of Things (IoT) devices and wearable health trackers to continuously monitor patient health parameters such as heart rate, temperature, and blood pressure. Based on this real-time data, automated alerts can be generated for users who require immediate medical attention or routine checkups.

Cloud computing can be implemented to ensure secure, scalable, and efficient storage of large volumes of healthcare data. Cloud-based systems will also allow seamless access to data from multiple healthcare centers, improving coordination between hospitals, clinics, and government health programs.

In addition, chatbot-based virtual assistants can be integrated into the system to provide instant responses to user queries regarding vaccination schedules, health tips, and appointment details. These AI-driven chatbots can operate 24/7, reducing the workload on healthcare staff.

Location-based services can further enhance the system by guiding users to nearby hospitals, clinics, and vaccination centers. This feature will be especially beneficial in emergency situations or during public health campaigns.

Finally, the system can be expanded to include emergency alert mechanisms for critical health conditions, ensuring faster response times and improved patient safety. With continuous advancements in mobile communication, artificial intelligence, and healthcare technology, this system has the potential to evolve into a comprehensive digital health management platform that significantly improves public health outcomes.

1. The system can be upgraded to support Artificial Intelligence-based prediction models to identify

users who are likely to miss vaccinations or health checkups.

2. Integration with national digital health platforms can enable automatic syncing of patient records and vaccination schedules in real time.
3. A multilingual voice call reminder system can be added for users who are illiterate or have difficulty reading SMS messages.
4. The system can be extended into a mobile application that provides additional features such as health reports, appointment booking, and doctor consultation.
5. IoT-based health monitoring devices can be integrated to automatically track patient health parameters and trigger alerts when necessary.
6. Cloud-based architecture can be implemented to improve scalability and ensure secure storage of large volumes of health data.
7. Chatbot support using AI can be introduced for answering basic health-related queries and guiding users.
8. GPS-based tracking can be used to identify nearby healthcare centers for vaccination and check-ups.
9. Integration with wearable devices can help in continuous health monitoring and personalized reminder generation.
10. The system can be expanded to include emergency alert services for critical health conditions.

# Chapter 5

# RECOMMENDATIONS AND CONCLUSION

## 5.1 Recommendations

In addition to the above recommendations, regular monitoring and evaluation should be conducted to measure the effectiveness of the SMS-based vaccination reminder system. Authorities should collect feedback from villagers, parents, and healthcare workers regarding message clarity, delivery timing, and usefulness of reminders. This feedback can help in identifying technical problems such as delayed messages, incorrect mobile numbers, or language barriers. By analyzing the responses, necessary improvements can be made to increase efficiency and user satisfaction. Continuous monitoring will ensure that the system remains reliable, accurate, and beneficial for the community over time. Furthermore, strong coordination between government health departments, local primary health centers, Anganwadi workers, ASHA workers, and village authorities is essential for the successful implementation of the system. Health workers should be trained to update vaccination schedules, register beneficiaries, and assist people who may have difficulty using mobile phones. Awareness programs, village meetings, and campaigns can also be organized to educate residents about the importance of timely vaccination and the benefits of reminder services. With proper planning, support, and community participation, the SMS-based vaccination reminder system can significantly improve vaccination coverage, reduce preventable diseases, and contribute to the overall health and well-being of Kaluvaya village. Based on the survey findings in Kaluvaya village, it is recommended to implement a pilot SMS-based vaccination reminder system at the local healthcare center to test its effectiveness in improving timely immunization. Beneficiary data, including mobile numbers and vaccination schedules, should be systematically recorded and maintained in a digital format. The SMS reminders should be sent in simple and local language to ensure better understanding and accessibility among rural populations. It is also recommended to conduct awareness programs to educate parents and guardians about the importance of vaccination and how SMS reminders can assist them. Healthcare workers should be trained to manage and update the system efficiently, ensuring accurate and timely communication. Additionally, collaboration with government health departments can help in scaling the solution to other villages. Continuous monitoring and evaluation should be carried out to assess

the system's impact and make necessary improvements for better performance and wider adoption.

1. Implement the SMS-based vaccination reminder system in Kaluvaya village to improve awareness and timely vaccination.
2. Send reminder messages in local languages such as Telugu for better understanding among villagers.
3. Schedule reminders multiple times before the vaccination date to reduce missed appointments.
4. Coordinate with local health workers and Anganwadi staff to maintain updated beneficiary records.
5. Improve mobile network connectivity in areas facing signal problems.
6. Conduct awareness programs to educate people about the benefits of vaccination and SMS reminders.
7. Ensure data privacy and secure handling of user mobile numbers and health records.
8. Expand the system to nearby villages after successful implementation in Kaluvaya village.
9. The system should support multiple regional languages to ensure accessibility for people in rural and semi-urban areas.
10. Integration with government health databases can improve accuracy in tracking vaccination schedules and patient records.
11. Automated scheduling should be enhanced to send reminders at optimal times, increasing the likelihood of user response.
12. The system should include two-way communication, allowing users to confirm appointments or request rescheduling via SMS.
13. Data privacy and security measures must be implemented to protect sensitive patient information.
14. Awareness campaigns should be conducted to educate people about the benefits of SMS-based health reminders.
15. The system can be extended to include reminders for other health services such as prenatal care, medication adherence, and routine screenings.
16. Incorporating AI-based analytics can help predict missed appointments and improve healthcare planning.

17. Collaboration with telecom providers can reduce SMS costs and improve delivery reliability.
18. A mobile application can be developed alongside SMS services for users with smartphones, offering additional features and better user interaction.

## 5.2 Conclusion

The survey conducted in Kaluvaya village shows that an SMS-based vaccination reminder system can effectively improve vaccination awareness and attendance. Most respondents expressed positive interest in receiving reminder messages and believed that the system would help reduce missed vaccination dates. Since mobile phone usage is high among villagers, SMS reminders are a practical and low-cost solution. Therefore, implementing this system can support public health initiatives, increase vaccination coverage, and contribute to the well-being of the community. In addition, the proposed system can strengthen communication between healthcare workers and residents by providing timely updates regarding vaccination camps, schedule changes, and important health announcements. This will help families stay informed and prepared, especially in rural areas where access to information may be limited.

Overall, the SMS-based vaccination reminder system is simple, reliable, and easy to use for people of all age groups. With proper implementation and regular monitoring, the system can become an effective tool for improving healthcare services and ensuring better protection against preventable diseases in Kaluvaya village and surrounding areas.

We would like to conclude that by emphasizing the opportunity that this Community service project has given us in solving real world problems.

This project underscores that even modest efforts, driven by passion, can catalyze enduring community transformation.

In conclusion, our community service project has yielded valuable insights and outcomes that highlight the power of collective action in creating positive change.

Through our efforts, we have witnessed the transformative effects of offering support, education, and resources to underserved members of our community. This project examined the feasibility of an SMS-based vaccination reminder system through a field survey conducted in Kaluvaya village. The findings revealed that while vaccination services are available, there are significant gaps in awareness, communication, and timely follow-up, leading to missed or delayed immunizations. The survey also showed that most households have access to mobile phones and are willing to receive SMS reminders, indicating strong potential for adopting a mobile-based solution. Based on the analysis, an SMS-based reminder system can serve as a simple, cost-effective, and practical approach to improve vaccination adherence in rural areas. It can help parents remember important dates, support healthcare workers in tracking beneficiaries, and enhance overall communication. Although this project did not involve the development of a prototype, it successfully highlights the importance and applicability of such a

system. Implementing this solution in the future can contribute to better immunization coverage and improved public health outcomes.

# Appendix A

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